

DATASHEET – EBASE7



DIMENSIONS and WEIGHT

Vehicle size	L = 1400 mm / 55.1 in	
	W = 820 mm / 32.2 in	
	H = 350 mm / 13.7 in	
Equipment	Integrated Lift platform	
	Plane size	L = 1300 mm / 51.1 in
		W = 720 mm / 28.3 in
	Stroke	Customisable depending on the payload
Vehicle weight w/o batteries	400 kg / 881.8 lbs	

PAYLOAD and REQUIREMENTS

Nominal payload 100% deliveries	1200 kg / 2645.5 lbs
Maximum payload 30% deliveries	1500 kg / 3306.9 lbs
Payload max size	L = 1200 mm / 47.2 in W = 1200 mm / 47.2 in H = 1500 mm / 59.0 in
Center of gravity footprint	L = 860 mm / 33.8 in W = 700 mm / 27.5 in
Center of gravity height	H = 800 mm / 31.4 in
Min load frame clearance	W = 1050 mm / 41.3 in H = 395 mm / 15.5 in

PERFORMANCE

Kinematics and Drive	Differential, bidirectional
Speed	0,1 m/s min / 0.2 mph 1,4 m/s max / 3.1 mph
X, Y accuracy	+/- 10 mm / 0.39 in
θ accuracy	+/- 1°
Localization rate	5 Hz
Minimum bending radius	0 m; the vehicle turns on its vertical axis
Slope	3% (1,7°)
Slope min bending radius	15 m / 0.59 in
Step	±5 mm / ±0.19 in
Gap	10 mm / 0.39 in
Motor wheel power	2x1500 W Brushless

Braking torque 2x5 Nm / 2 x 3.68 lb-ft

MATERIALS

Wheels coating Polyurethane grip + Vulkollan®

Chassis material AISI 304

Shell material Painted Steel

BATTERY

Battery LiFePO4 24V 100 Ah

Depth of discharge 70%

Autonomy 6 h

Recharging time 2.5 h

Battery weight 38 kg / 83.77 lbs

Battery cycle life 2.000 cycles

COMMUNICATION

Communication WI-FI - WLAN 802.11a/b/g/n - 2,4/5 GHz

SAFETY

Safety devices Nr. 2 Sick MicroScan3 Pro laser scanners
Sensitive board (front/rear)
N.2 E-stop buttons
Safety control central unit Sick Flexi EFlpro

Performance Level d

Safety field shape 360°

NAVIGATION

Navigation	Odometric closed loop via laser scanner (natural navigation)
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Mapping	Auto acquisition by Mobile Robot
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ENVIRONMENT

Temperature	from +5 to 35 °C / + 41 to 95 F
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Working relative humidity	from 30% to 80% (absence of condensation)
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IP degree	13
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REGULATIONS

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|--------------------|---|
| Regulations | • 2006/42/EC Machinery Directive |
| | • 2014/35/EU Low Voltage Directive |
| | • 2014/30/EU Electromagnetic Compatibility Directive |
| | • ISO 3691-4: 2023 Safety of driverless industrial trucks |